Wenyu Jiao

+86 15650709679 astrojiao@gmail.com wenyujiao@github.io

Research Interests

- Interferometric observations toward the cradle of high-mass stars
- Magnetic fields and dust polarization in star-forming regions
- Fragmentation and chemical evolution in infrared dark clouds
- Multi-scale gas flows associated with star formation activities

Education

2018 – 2024 **Peking University (PKU)**
Ph.D. candidate in Astrophysic. Supervisor: Prof. Ke Wang
Thesis: Exploring the Initial Conditions in High-mass Star Formation

2014 – 2018 **Peking University (PKU)**
B.S. in Astronomy. Supervisor: Prof. Ran Wang
Thesis: Submillimeter Study of A Giant Protocluster at z=5.7

Research Experience

2024.07 – now **Postdoc**
Shanghai Astronomical Observatory (SHAO). Collarborators: Prof. Tie Liu, Prof. Xing Lu

2023.02 – 2024.02 **Guest Researcher**
Max Planck Institute for Astronomy (MPIA). Supervisor: Prof. Henrik Beuther
Investigating the role of magnetic fields in high-mass star formation using ALMA high-resolution dust polarization data.

2022.08 – 2023.02 **Visitor**
Yunnan University (YNU)

Mentoring Experience

2024–now Co-mentor of two graduate students (Yang, Yuhan and Jiao, Yunfan) at Shanghai Astronomical Observatory (SHAO)

2021 Co-mentor of National Talent Program for high school student (Xie, Jiahe): "Exploring the relation between $^{13}CO/C^{18}O$ and H_2 in the galactic plane", current: Bachelor student at Beihang University

Mentoring Experience (continued)

2021 Co-mentor of undergraduate project (Huang, Haomiao): "The initial stage of star formation: Observational study of dense cloud cores with extremely low luminosity/mass ratio", current: Ph.D. student in National Astronomical Observatories, Chinese Academy of Sciences (NAOC, China)

2020 Co-mentor of a visiting student (Huang, Bo): "High-mass Starless Clumps: Dynamical State and Correlation between Physical Parameters", current: Postdoc at Korea Astronomy and Space Science Institute (KASI, South Korea)

Honors, Awards and Grants

2024 **Shanghai Post-doctoral Excellence Program**, Shanghai

2022 **China Scholarship Council fellowship**, China

2021 **Excellent Research Award**, Peking University

2019 **Presidential Scholarship**, Peking University

2019 **Canon Scholarship**, Peking University

2019 **Merit Student**, Peking University

Telescope Observing Projects

- "Probing the Initial conditions of extremely cold and low luminosity-to-mass ratio high-mass starless clumps within 3.5 kpc", SMA, 2021B-S032, **PI**, 3 tracks allocated, 2 tracks observed.
- "Magnetic Fields in Supercritical Massive Filaments", ALMA, 2025.1.00883.S, **Co-PI**
- "Penetrating the "thickness": high-resolution 3mm survey unveil REAL core mass in massive clumps", ALMA, 2025.1.00542.S, PI: Siju Zhang, Co-I
- "What is the importance of the Magnetic Field in the Formation of High-mass Stars?", ALMA, 2025.1.01549.S, PI: Patricio Sanhueza, Co-I
- "Panoramic view of how massive protoclusters form in hub-filament systems", ALMA, 2025.1.00044.S, PI: Guido Garay, Co-I
- "The origin of the wide-spread shocks in the Central Molecular Zone: excitation conditions in outflows and slim filaments", ALMA, 2024.1.00213.S, PI: Junhao Liu, Co-I
- "Magnetic Fields in Massive Collapsing Clumps", ALMA, 2023.1.01606.S, PI: Lei Zhu, Co-I
- "Tracing the core mass growth in filaments", NOEMA, PI: Shanghuo Li, Co-I
- "How really massive are protostellar cores? Determining the cm SED by ATCA", ATCA, PI: Fengwei Xu, Co-I

Selected Talks

2025 **Perspective in Star Formation—Senior Insights, Young Visions**, Shanghai, China
"From ATOMS to QUARKS: Unveiling the Mechanisms of Massive Star Formation with ALMA", Shanghai, Selected talk

JCMT Users Meeting, Chiang Mai, Thailand
"SPACE: Submillimeter Polarization and Astro-Chemistry in Earliest star formation", Selected talk

2024 **Academic Symposium on Molecular Clouds and Star Formation**, Guangxi, China
"The Role of Magnetic Fields in G335.579-0.292", Selected talk

Annual Meeting of the Chinese Astronomical Society, Zhejiang, China
"Gravitational Shaping of Magnetic Fields and Their Interplay with Gas Streamers in G335.579-0.292", Selected talk

2022 **Academic Symposium on Molecular Clouds and Star Formation** (remote)
"Are There High-mass Prestellar Cores in Orion?", Selected talk

2022 **CASSACA ALMA user meeting** (remote)
"Exploring the Initial Conditions in High-Mass Star Formation Using ALMA Data", Selected talk

Services

2025	LOC Chair of "Perspectives in Star Formation — Senior Insights, Young Visions" Workshop in Shanghai with 140 participants
2021	Organizer of PKU Undergraduate Astronomy Symposium
2021	Teaching assistant of Modern Astronomy Course, Peking University
2020	Teaching assistant of ALMA Data Reduction Workshop, Yunnan Province, China
2020	Operating staff for Official WeChat Public Account of "Peking University Astronomy"
2019-2022	Mentor of the 2019 DoA undergraduate class, School of Physics, Peking University

Skills

Software	CASA, CARTA, RADMC-3D, RADEX, GILDAS, $getsf$, DS9
Analysis	ALMA, IRAM 30m, SMA, JCMT, APEX, Herschel, Planck, Nobeyama 45m

References

Prof. Tie Liu
Shanghai Astronomical Observatory
Email: liutie@shao.ac.cn

Prof. Ke Wang

The Kavli Institute for Astronomy and Astrophysics at Peking University
Email: kwang.astro@pku.edu.cn

Prof. Henrik Beuther
Max Planck Institute for Astronomy
Email: beuther@mpia.de

Publications

Refereed Publications (3 first author + 28 co-authored publications) . . . . . .

- **Jiao, W.**; Wang, K.; Pillai, T. G.; and 3 coauthors, "Fragmentation of the High-mass 'Starless' Core G10.21-0.31: A Coherent Evolutionary Picture for Star Formation", 2023, ApJ, 945, 81
- **Jiao, W.**; Wang, K.; Xu, F.; and 2 coauthors, "Relative Alignment between Gas Structures and Magnetic Field in Orion A at Different Scales using Different Molecular Gas Tracers", 2024, A&A, 686, A202
- **Jiao, W.**; Wang, K.; Xu, F., "Absence of High-mass Prestellar Cores in the Orion Giant Molecular Cloud", 2024, AJ, 168, 151
- Huang, B.; Wang, K.; Girart, J. M.; **Jiao, W.**, and 2 coauthors, "High-mass Starless Clumps: Dynamical State and Correlation between Physical Parameters", 2023, ApJ, 949, 46
- Xu, F.; Wang, K.; Schneider, N.;...**Jiao, W.**; et al., "Linear filament and nested cluster evolution tomography (LANCET) I. Capture the evolution of dense gas in 14-parsec filament G316.8", 2026, A&A, in press
- Cai, K.; Huang, B.; Girart, J.M.;...**Jiao, W.**; et al., "B-field Orion Protostellar Survey (BOPS). IV: The Relative Orientation Between Magnetic Fields and Density Structures in Young Protostellar Envelopes", 2026, ApJ, in press
- Gupta, S.; Baug, T.; Soam, A.;...**Jiao, W.**; et al., "ATOMS-QUARKS survey: Inflow and infall in massive protocluster G318.049+00.086: Evidence of competitive accretion", 2026, ApJ, in press
- Liu, Y.; Lu, X.; Liu, J.;... **Jiao, W.**; et al., "ALMA Polarization Study of the Magnetic Fields in Two Massive Clumps in the 20 km s1 Cloud of the Central Molecular Zone", 2026, ApJ, 998, 330
- Meng, D.; Liu, T.; Esimbek, J.;... **Jiao, W.**; et al., "The ALMA-QUARKS Survey: Hot Molecular Cores Are a Long-standing Phenomenon in the Evolution of Massive Protostars", 2026, ApJ, 997, 340
- Zhang, S.; Garay, G.; Xu, F.;... **Jiao, W.**; et al., "ALMA-QUARKS: Few-thousand-year "Hatching out of the Egg:" The Supersonic Breakout of a Hypercompact H II Region from Its Parental Hot Core", 2026, ApJ, 997, 201
- Hwang, J.; Sanhueza, P.; Girart, J. M.;... **Jiao, W.**; et al., "Magnetic Fields in Massive Star-forming Regions (MagMaR). VI. Magnetic Field Dragging in the Filamentary High-mass Star-forming Region G35.200.74N Due to Gravity", 2026, AJ, 171, 50

- Yang, D.; Liu, H.; Qin, S.;... **Jiao, W.**; et al., "The ALMA-QUARKS Survey: Evidence of a Candidate High-mass Prestellar Core Aside a Bright-rimmed Cloud IRAS 18290-0924", 2025, ApJ, 995, 193
- Mai, X.; Liu, T.; Liu, X.;... **Jiao, W.**; et al., "A misaligned protostellar disk fed by gas streamers in a barred spiral-like massive dense core", 2025, SciAdv, 11, eady6953
- Yang, D.; Liu, H.; Liu, T.;... **Jiao, W.**; et al., "The ALMA-QUARKS Survey. III. Clump-to-core Fragmentation and Searches for High-mass Starless Cores", 2025, ApJS, 280, 33
- Issac, N.; Lu, X.; Liu, T.;... **Jiao, W.**; et al., "Detection of an Explosive Outflow in G34.26+0.15", 2025, AJ, 169, 324
- Li, Z.; Liu, X.; Liu, T.;... **Jiao, W.**; et al., "The ALMA-ATOMS survey: A sample of weak hot core candidates identified through line stacking", 2025, A&A, 697, A190
- Xu, F.; Lu, X.; Wang, K.;... **Jiao, W.**; et al., "Dual-band Unified Exploration of three CMZ Clouds (DUET): Cloud-wide census of continuum sources showing low spectral indices", 2025, A&A, 697, A164
- Liu, X.; Liu, T.; Mai, X.;... **Jiao, W.**; et al., "Core Mass Function in View of Fractal and Turbulent Filaments and Fibers", 2025, RAA, 25, 025020
- Sanhueza, P.; Liu, J.; Morii, K.;... **Jiao, W.**; et al., "Magnetic Fields in Massive Star-forming Regions (MagMaR). V. The Magnetic Field at the Onset of High-mass Star Formation", 2025, ApJ, 980, 87
- Zhang, Z.; Lu, X.; Liu, T.;... **Jiao, W.**; et al., "ALMA Observations of Massive Clouds in the Central Molecular Zone: External-pressure-confined Dense Cores and Salpeter-like Core Mass Functions", 2025, ApJ, 980, 44
- Chen, L.; Qin, S.; Liu, T.;... **Jiao, W.**; et al., "The ALMA-ATOMS survey: Vibrationally excited HC3N lines in hot cores", 2025, A&A, 694, A166
- Yang, D.; Liu, H.; Liu, T.;... **Jiao, W.**; et al., "The ALMA-QUARKS Survey: Fibers' Role in Star Formation Unveiled in an Intermediate-mass Protocluster Region of the Vela D Cloud", 2024, ApJ, 976, 241
- Zhang, C.; Liu, T.; Ren, Z.;... **Jiao, W.**; et al., "ATOMS: ALMA three-millimetre observations of massive star-forming regions - XVI. Neutral versus ion line widths", 2024, MNRAS, 533, 4234
- Zapata, L.; Fernández-López, M.; Sanhueza, P.;... **Jiao, W.**; et al., "Magnetic Fields in Massive Star-forming Regions (MagMaR). IV. Tracing the Magnetic Fields in the O-type Protostellar System IRAS 16547–4247", 2024, ApJ, 974, 257
- Shen, X.; Liu, H.; Ren, Z.;... **Jiao, W.**; et al., "JCMT 850 m Continuum Observations of Density Structures in the G35 Molecular Complex", 2024, ApJ, 974, 239
- Saha, P.; Sanhueza, P.; Padovani, M.;... **Jiao, W.**; et al., "Magnetic Fields in Massive Star-forming Regions (MagMaR): Unveiling an Hourglass Magnetic Field in G333.460.16 Using ALMA", 2024, ApJL, 972, L6
- Cortés, P.; Girart, J. M.; Sanhueza, P.;... **Jiao, W.**; et al., "MagMaR III—Resisting the Pressure, Is the Magnetic Field Overwhelmed in NGC6334I?", 2024, ApJ, 972, 115

- Xu, F.; Wang, K.; Liu, T.;... **Jiao, W.**; et al., "The ALMA Survey of Star Formation and Evolution in Massive Protoclusters with Blue Profiles (ASSEMBLE): Core Growth, Cluster Contraction, and Primordial Mass Segregation", 2024, ApJS, 270, 9
- Mai, X.; Liu, T., Liu, X.,... **Jiao, W.**; et al., "The ALMA-QUARKS survey: Detection of two extremely dense substructures in a massive prestellar core", 2024, ApJL, 961, L35
- Xu, F.; Wang, K.; Liu, T.; ... **Jiao, W.**; et al., "ATOMS: ALMA Three-millimeter Observations of Massive Star-forming regions - XV. Steady accretion from global collapse to core feeding in massive hub-filament system SDC335", 2023, MNRAS, 520, 3259
- Zhang, S.; Wang, K.; Liu, T.;... **Jiao, W.**; et al., "ATOMS: ALMA three-millimeter observations of massive star-forming regions - XIII. Ongoing triggered star formation within clump-fed scenario found in the massive (1500 $M_\odot$) clump", 2023, MNRAS, 520, 322

Publications submitted

- Zhang, Y.; Liu, T.; **Jiao, W.**; et al., "The ALMA-QUARKS Survey: Discovery of Dusty Fibrils inside Massive Star-forming Clumps", submitted to AJ.
- Jiao, Y.; Liu, T.; **Jiao, W.**; et al., "The ALMA-QUARKS survey: Multipolar episodic molecular outflow associated with the most energetic water maser source in the Galaxy", submitted to AJ.
- Yang, Y.; Liu, T.,; Liu, S.; ...**Jiao, W.**; et al., "Tracking Protostellar Variability in Massive Protoclusters with ALMA: I. Insights from QUARKS and MaMMOtH", submitted to ApJ.
- Lee, G.; Kim, K.,; Liu, T.; ... **Jiao, W.**; et al., "The ALMA-QUARKS survey: A Keplerian Circumbinary Disk Candidate in the High-mass Star-forming Region IRAS 08303–4303", submitted to ApJ.
- Yang, D.; Liu, H.,; Qin, S.;... **Jiao, W.**; et al., "Evolution of starless cores in massive clumps seen by the ALMA ASHES and QUARKS surveys", submitted to ApJ.